

Build A Pose Map

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2. Building a Grid Map

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1. Use slam_toolbox to build grid maps and pose maps

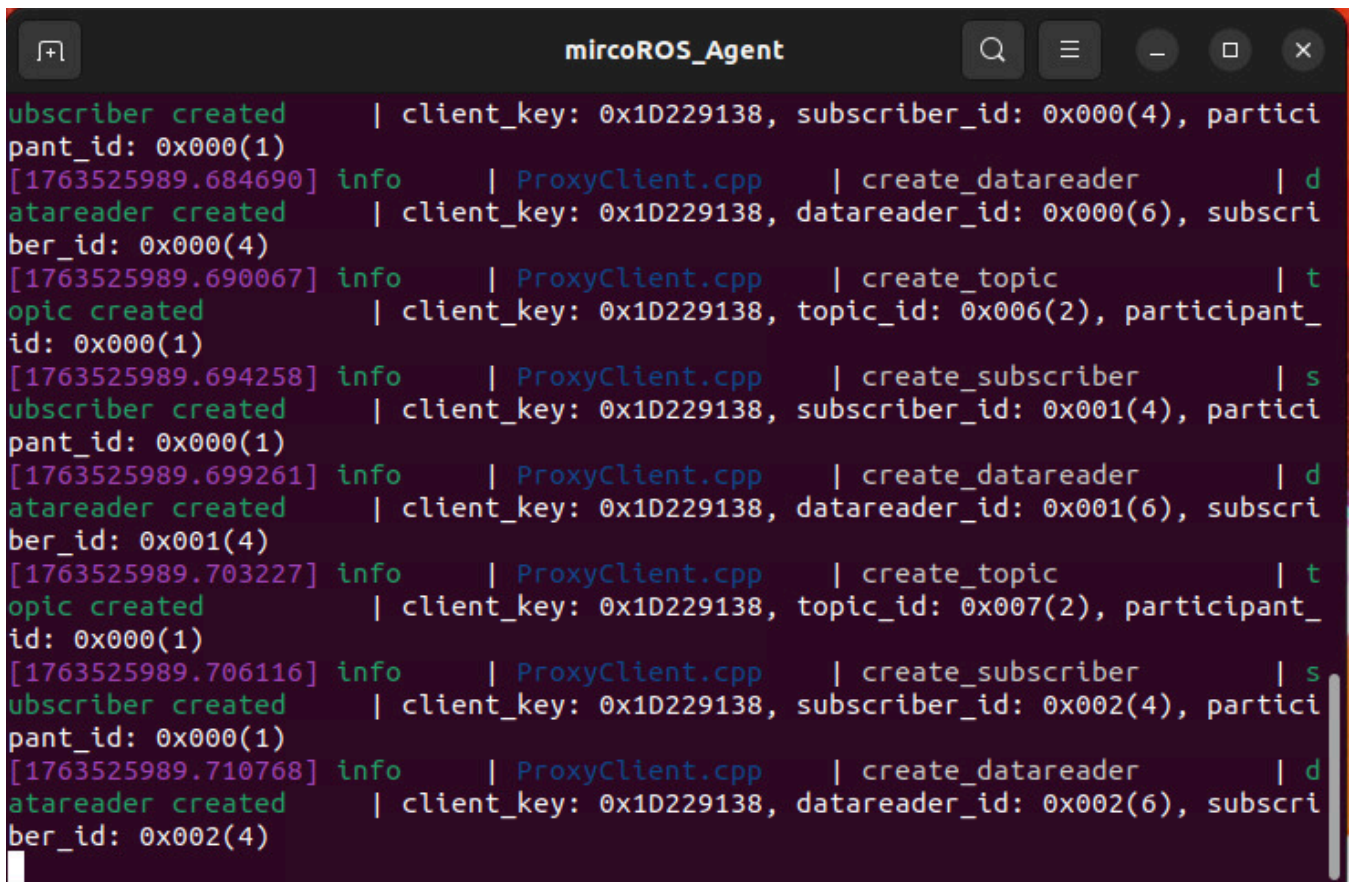
Note

- At least one grid map is required before marking the road network .geojson file
- The pose map enables fast relocalization when marking waypoints and navigating, improving global localization accuracy and enhancing waypoint marking accuracy

2. Building a Grid Map

- If the chassis mircoROS agent communication is not started, first start the agent. If already started, no need to restart

```
sh start_agent.sh
```



```
subscriber created      | client_key: 0x1D229138, subscriber_id: 0x000(4), partici
pant_id: 0x000(1)
[1763525989.684690] info      | ProxyClient.cpp      | create_datareader      | d
atareader created      | client_key: 0x1D229138, datareader_id: 0x000(6), subscri
ber_id: 0x000(4)
[1763525989.690067] info      | ProxyClient.cpp      | create_topic            | t
opic created           | client_key: 0x1D229138, topic_id: 0x006(2), participant_
id: 0x000(1)
[1763525989.694258] info      | ProxyClient.cpp      | create_subscriber       | s
subscriber created     | client_key: 0x1D229138, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1763525989.699261] info      | ProxyClient.cpp      | create_datareader       | d
atareader created     | client_key: 0x1D229138, datareader_id: 0x001(6), subscri
ber_id: 0x001(4)
[1763525989.703227] info      | ProxyClient.cpp      | create_topic            | t
opic created           | client_key: 0x1D229138, topic_id: 0x007(2), participant_
id: 0x000(1)
[1763525989.706116] info      | ProxyClient.cpp      | create_subscriber       | s
subscriber created     | client_key: 0x1D229138, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1763525989.710768] info      | ProxyClient.cpp      | create_datareader       | d
atareader created     | client_key: 0x1D229138, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
```

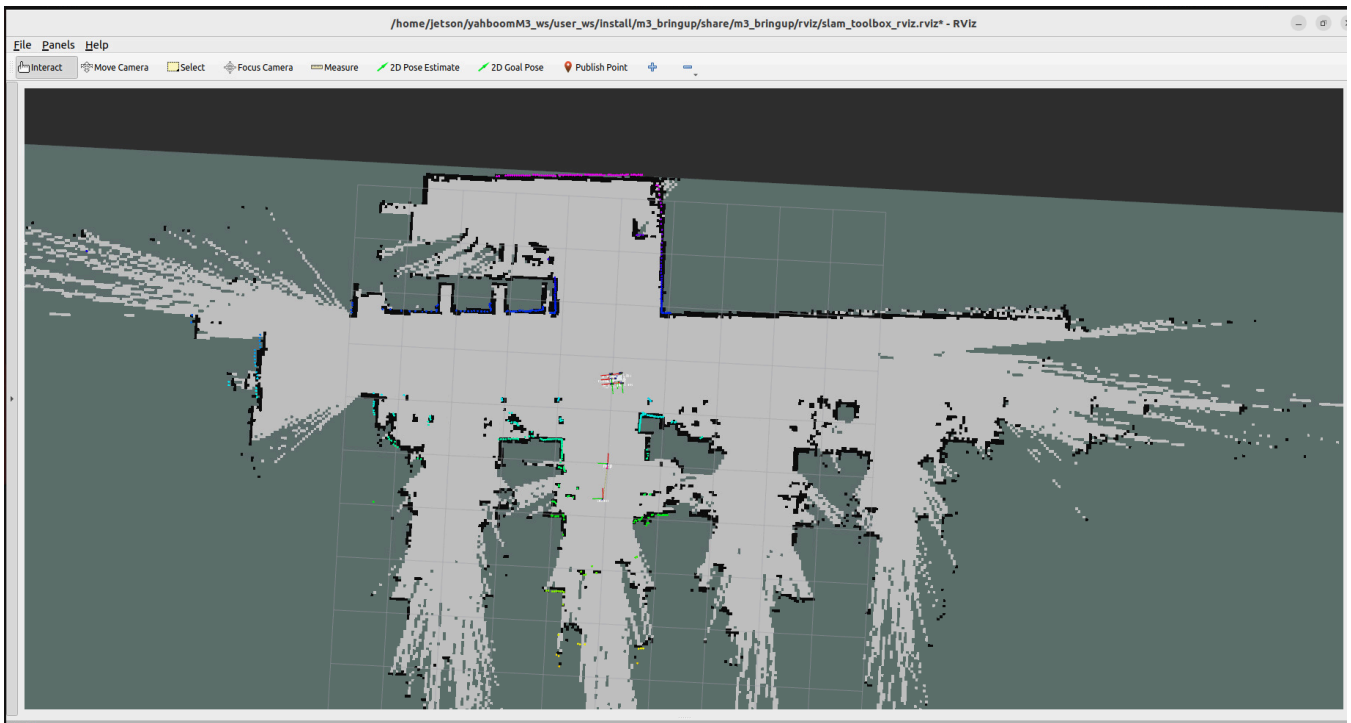
- Enter the command in the terminal

```
ros2 launch slam_mapping slam_toolbox.launch.py
```

- Then start the keyboard control node (you can also use a gamepad)

```
ros2 run teleop_twist_keyboard teleop_twist_keyboard
```

- Control the robot movement for mapping



- Save the Grid Map

```
ros2 launch slam_mapping save_slamtoolbox_map.launch.py
```

💡 Tip

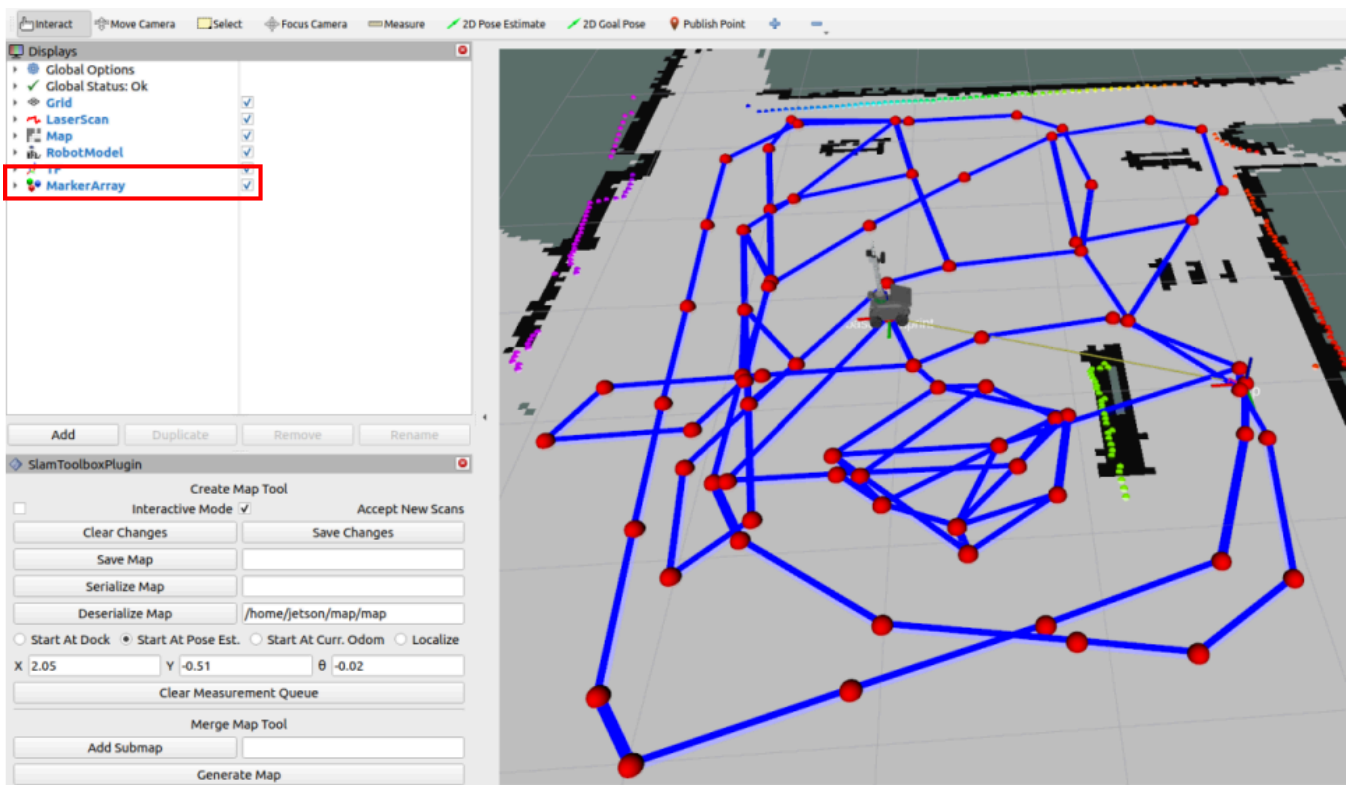
- `map_name` is used to specify the **map name** of the grid map to be saved. If no launch parameter is added, the default map name is `map`, and all map files are saved in the `~/map` directory,

The terminal prompt `Map saved successfully` confirms the map was saved successfully

```
jetson@yahboom: ~ 82x22
[map_saver_cli-1] map_saver lifecycle node launched.
[map_saver_cli-1] Waiting on external lifecycle transitions to activate
[map_saver_cli-1] See https://design.ros2.org/articles/node_lifecycle.html f
or more information.
[map_saver_cli-1] [INFO] [1763532975.329685271] [map_saver]: Creating
[map_saver_cli-1] [INFO] [1763532975.330069728] [map_saver]: Configuring
[map_saver_cli-1] [INFO] [1763532975.341867665] [map_saver]: Saving map from 'map'
topic to '/home/jetson/map/map' file
[map_saver_cli-1] [WARN] [1763532975.359932772] [map_io]: Image format unspecified
. Setting it to: pgm
[map_saver_cli-1] [INFO] [1763532975.361316136] [map_io]: Received a 224 X 598 map
@ 0.05 m/pix
[map_saver_cli-1] [INFO] [1763532975.439369065] [map_io]: Writing map occupancy da
ta to /home/jetson/map/map.pgm
[map_saver_cli-1] [INFO] [1763532975.443013992] [map_io]: Writing map metadata to
/home/jetson/map/map.yaml
[map_saver_cli-1] [INFO] [1763532975.444847063] [map_io]: Map saved
[map_saver_cli-1] [INFO] [1763532975.446514786] [map_saver]: Map saved successfull
y
[map_saver_cli-1] [INFO] [1763532975.449017438] [map_saver]: Destroying
[INFO] [map_saver_cli-1]: process has finished cleanly [pid 50218]
jetson@yahboom:~$
```

- Check pose point quantity (Important)

- Click the MarkerArray option in the Display panel of Rviz to view the pose points of the pose map. Ensure there are sufficient pose points in the map area. If there are too few or no pose points in a certain area, the positioning effect in that area during navigation will be relatively poor.



- Save the Pose Map

```
ros2 launch slam_mapping save_slamtoolbox_map.launch.py map_type:=posegraph
```

💡 Tip

- `map_type` Save map type, defaults to `gridmap` for saving grid maps. When `posegraph` is selected, the `slam_toolbox` pose map is saved

The terminal prompt `slam_toolbox.srv.SerializePoseGraph_Response(result=0)` confirms the pose map has been saved

```
jetson@yahboom: ~ 82x22
jetson@yahboom:~$ ros2 launch m3_bringup save_map.launch.py map_name:=/home/jetson
/map/map map_type:=posegraph
[INFO] [launch]: All log files can be found below /home/jetson/.ros/log/2025-11-19
-14-19-41-297598-yahboom-66778
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [ros2-1]: process started with pid [67023]
[ros2-1] requester: making request: slam_toolbox.srv.SerializePoseGraph_Request(fi
lename='/home/jetson/map/map')
[ros2-1]
[ros2-1] response:
[ros2-1] slam_toolbox.srv.SerializePoseGraph_Response(result=0)
[ros2-1]
[INFO] [ros2-1]: process has finished cleanly [pid 67023]
jetson@yahboom:~$
```

- Map files are saved in the `~/map` directory by default

